Arkaprava Majumdar

ai24mtech02002@iith.ac.in | ai24mtech02002@iith.ac.in | +91 7044942883 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.26
Modern English Academy, Barrackpore	Class XII (Council for the Indian School Certificate Examinations)	2015	93.17%
Modern English Academy, Barrackpore	Class X (Council for the Indian School Certificate Examinations)	2013	94.33%

THESIS

Neural Wave Equation for Continuous-Depth Dynamics

- Developed a **continuous-depth neural architecture** that evolves latent states through a non-homogeneous wave PDE, using adaptive numerical integration for sequence and dynamical-system modelling.
- Evaluated **solver behaviour, runtime, memory, and predictive accuracy** across human activity recognition, sepsis prediction, and physical-system forecasting tasks.

Surrogate PDE Dynamics

- Built and evaluated **learned surrogate models for time-dependent PDEs**, comparing one-step predictors, continuous-time flows, and structure-aware dynamics across multiple systems.
- Designed long-horizon rollout evaluation covering **trajectory accuracy, stability, conservation-law violations, NFE, and wall-clock cost**, highlighting trade-offs between predictive performance and physical consistency.

PUBLICATION

Neural Wave Equation for Irregularly Sampled Sequence Data

International Conference on Learning Representations (ICLR), 2025

- Proposed the **Neural Wave Equation**, a neural PDE architecture for modelling irregularly sampled sequential data.
- Modelled latent-state evolution **continuously across both time and network depth** using a non-homogeneous wave equation parameterised by a neural network.
- Demonstrated improved performance across irregularly sampled sequence-labelling tasks, with theoretical analysis connecting the wave formulation to denser hidden-state connectivity.

Bayesian Calibration of the Heston Model Across Physical and Risk-Neutral Measures

FinDS 2026 -- Workshop on Data Management for Modern Financial Systems, co-located with SIGMOD/PODS 2026 | Short Paper, Oral Presentation

- Developed a **Bayesian joint-calibration framework** for the Heston stochastic-volatility model using information from both historical returns and option-price surfaces.
- Combined **particle filtering and likelihood-based inference** to jointly estimate latent volatility, model parameters, and the volatility risk premium across physical and risk-neutral measures.
- Addressed parameter **non-identifiability in conventional single-measure calibration**, demonstrating more consistent parameter recovery through joint inference.

Not All That's Colorful Is Real: Rethinking Metrics for Image Colorization

NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation (GenProCC)

- Investigated limitations of conventional reference-based and distribution-level metrics for evaluating the inherently ambiguous task of image colourization.
- Benchmarked **20+ no-reference image-quality and colourfulness metrics** across three datasets and more than **100,000 generated images**.
- Developed a **rank-based evaluation framework** and showed that no-reference IQA metrics such as HyperIQA and TOPIQ provide effective per-image assessments of colourisation realism.

ISA-Bench: A Benchmark for Computational Reasoning Across Instruction Set Architectures

MLForSys Workshop Neurips 2026

- Introduced a benchmark for evaluating **LLM computational reasoning under unfamiliar and constrained instruction-set architectures**, beyond conventional Python/Java code-generation benchmarks.
- Built complete execution environments including **parsers, virtual machines, verifiers, and structured-feedback loops** for automated single-shot and iterative evaluation.
- Introduced the **Reasoning-Execution Gap (REG)** analysis, identifying failures where models infer an appropriate computational strategy but cannot correctly express it in the target instruction set.

PROJECTS

Code Inference Scaling - LLM Systems & Evaluation Pipeline

Python · vLLM · Qwen3 · Multiprocessing · Static Analysis

- Built an end-to-end **LLM code-generation evaluation pipeline** spanning benchmark preprocessing, vLLM serving, solution generation, isolated correctness testing, and performance analysis.
- Evaluated **86K+ generations** across four Qwen3 model sizes and five reasoning-budget configurations, studying how model scale, inference budget, and code complexity affect pass@1.
- Developed a **bounded multiprocessing test harness** and AST/complexity analysis pipeline to safely evaluate generated programs and correlate structural code complexity with inference behaviour.

Mini-SGLang DRY Sampling - LLM Inference Systems

Python · Triton · GPU Kernels · LLM Inference

- Implemented **DRY (Don't Repeat Yourself) sampling** inside a Mini-SGLang inference pipeline to penalise repeated token sequences during generation.
- Developed a **Triton GPU kernel** using windowed sequence-matching logic to identify repetitions and apply token-level penalties efficiently during decoding.
- Integrated the sampler with **batched inference and sampling infrastructure**, profiling correctness and performance across GPU and reference implementations.

SlotCore -- Agentic Slot-Game Generation Framework

Bash · C++ · Python · LLM Agents · Clang/AST

- Built an **agentic software-generation pipeline** for creating slot-game implementations from high-level specifications, with separate planning, implementation, validation, and retry stages.
- Integrated **compiler-assisted verification using Clang/AST tooling** to inspect generated C++ code and enforce structural and mathematical constraints.
- Designed automated **simulation and validation loops** to test generated games for correctness, payout behavior, and consistency with target mathematical specifications.

WORK EXPERIENCE

AI Research Intern, Pi School

04/12/2023 - 09/02/2024 (Remote Internship, Rome, Italy)

- Built an **end-to-end PDF-to-LLM evaluation pipeline** for academic papers, including document parsing, text extraction, chunking, prompt construction, and automated question answering.
- Designed and evaluated **chunking and context-selection strategies** for answering predefined statistical and methodological questions over long research papers.

Junior Research Fellow, Indian Institute of Science Education and Research (IISER-Berhampur)

24/08/2022 - 05/07/2023 (Research Fellowship)

- Applied statistical learning and explainable AI methods to MRI and radiogenomic datasets for neurological disease analysis.

- Co-authored a poster on the same project, accepted at the International Society of Magnetic Resonance in Medicine, 2023.

Slot Game Mathematician, Bonanza Interactive

01/08/2020 - 31/12/2020(Internship), 4/01/2021 - 23/08/2022 (Full Time)

- Built large-scale **Monte Carlo simulation frameworks** for slot-game mathematics, supporting **RTP tuning, volatility analysis, hit-frequency estimation, and game balancing** across production game configurations.
- Developed and validated **probabilistic models and RNG-driven game mechanics**, using simulation-based statistical analysis to verify mathematical specifications, payout behavior, and game performance.

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award (IIT-H)

Research Appreciation Award (IIT-H)

SKILLS

Programming Languages: C++, Python,Triton,Bash,SQL

Deep Learning Frameworks: Jax, Pytorch, TensorFlow

Statistics, Probability, Stochastic Processes

RELEVANT COURSES

Foundations of Machine Learning, Deep Learning, Probability and Stochastic Process, Matrix Theory, Bayesian Data Analysis, Stochastic Algorithms for Nonconvex Optimization and Reinforcement Learning, Introduction to Stochastic Differential Equations

EXTRACURRICULAR

Amazon ML Challenge, Top 50 Finish; Team Rank 19 in 2025 and Team Rank 44 in 2024

IMC Prosperity 4 Trading Competition, Manual Trading Round Rank 676, Team Rank 1219

POSITIONS OF RESPONSIBILITY

Placement Cell Coordinator (AI Depratment, IIT-H)